RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 2C: Vectors III – Equations of Planes (Solutions)

1 9740/N2007/1/Q8

The line l passes through the points A and B with coordinates $(1,2,4)$ and $(-2,3,1)$ respectively. The plane p has equation $3x-y+2z=17$. Find

(i) the coordinates of the point of intersection of l and p, [5]

(ii) the acute angle between l and p, [3]

(iii) the perpendicular distance from A to p. [3]

(i) $\overrightarrow{AB}=\begin{pmatrix}-3\\1\\-3\end{pmatrix}$

$l:\ \mathbf{r}=\begin{pmatrix}1\\2\\4\end{pmatrix}+\lambda\begin{pmatrix}-3\\1\\-3\end{pmatrix}, \lambda\in\mathbb{R}$

$p:\ \mathbf{r}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=17$

When l meets p at point C,

$$\left[\begin{pmatrix}1\\2\\4\end{pmatrix}+\lambda\begin{pmatrix}-3\\1\\-3\end{pmatrix}\right]\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=17$$

$$(3-2+8)+\lambda(-9-1-6)=17$$

$$\lambda=-\frac{1}{2}$$

$$\overrightarrow{OC}=\begin{pmatrix}1\\2\\4\end{pmatrix}-\frac{1}{2}\begin{pmatrix}-3\\1\\-3\end{pmatrix}=\begin{pmatrix}\frac{5}{2}\\\frac{3}{2}\\\frac{11}{2}\end{pmatrix}$$

The coordinates of C are $\left(\frac{5}{2},\frac{3}{2},\frac{11}{2}\right)$.

(ii)	Let θ be the angle between the line and the normal to plane. $\cos\theta = \dfrac{\begin{pmatrix}3\\-1\\3\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}}{\sqrt{19}\sqrt{14}} = \dfrac{16}{\sqrt{19}\sqrt{14}} \quad\Rightarrow\quad \theta = 11.2^{\circ}\,(1\text{dp})$ Hence required angle is $90^{\circ} - 11.2^{\circ} = 78.8^{\circ}$ Alternative: Let α be the angle between *l* and *p*. $\sin\alpha = \dfrac{\left|\begin{pmatrix}3\\-1\\3\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}\right|}{\sqrt{19}\sqrt{14}} = \dfrac{16}{\sqrt{19}\sqrt{14}} \quad\Rightarrow\quad \theta \approx 78.820^{\circ} \approx 78.8^{\circ}$
(iii)	[We know that $C\ \left(\dfrac{5}{2},\dfrac{3}{2},\dfrac{11}{2}\right)$ lies on p.] $\overrightarrow{AC} = -\dfrac{1}{2}\begin{pmatrix}-3\\1\\-3\end{pmatrix}$ Method 1: Distance is $AC\sin\theta = \dfrac{1}{2}\sqrt{3^2+1+3^2}\,\sin 78.820 \approx 2.14$ Method 2: Distance is $\left|\overrightarrow{AC}\cdot\hat{\mathbf{n}}\right| = \dfrac{\left|-\dfrac{1}{2}\begin{pmatrix}-3\\1\\-3\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}\right|}{\sqrt{3^2+1^2+2^2}}$ $= \dfrac{\lvert -9-1-6\rvert}{2\sqrt{14}} = \dfrac{8}{\sqrt{14}}$ Method 3: Distance is $\dfrac{\left|\begin{pmatrix}1\\2\\4\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}-17\right|}{\sqrt{3^2+1^2+2^2}} = \dfrac{\lvert 9-17\rvert}{\sqrt{14}} = \dfrac{8}{\sqrt{14}}$

2 AJCPrelim9740/2010/1/Q13

The equations of a plane π_1 and a line l are shown below:

$$\pi_1 : \mathbf{r}.\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 6$$

$$\frac{x-1}{3} = \frac{y+4}{2} = z-5$$

The point A has position vector $3\mathbf{i} - \mathbf{j} + 4\mathbf{k}$.

(i) Find the distance between point A and the plane π_1. [2]

(ii) B is another point such that $\overrightarrow{AB} = -5\mathbf{j} - 2\mathbf{k}$. Find the length of projection of $\overrightarrow{AB}$ onto the plane π_1. [2]

(iii) Using your answers in **(i)** and **(ii)**, find the area of triangle ABC, where C is the reflection of A in the plane π_1. [2]

(iv) Find the equation of the plane π_2 which contains the line l and the origin. Hence, find the line of intersection between the planes, π_1 and π_2. [4]

(i) The point D with $\overrightarrow{OD} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$ lies in π_1.

Distance is $\left|\overrightarrow{DA} \cdot \hat{\mathbf{n}}\right| = \dfrac{\left|\left(\overrightarrow{OA} - \overrightarrow{OD}\right) \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\right|}{\sqrt{2^2 + 1^2 + 2^2}}$

$$= \frac{\left|\begin{pmatrix} 0 \\ -1 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\right|}{\sqrt{2^2 + 1^2 + 2^2}}$$

$$= \frac{|-1-8|}{3} = 3$$

OR

Let P be any point on π_1.

Distance is $\left|\overrightarrow{PA} \cdot \hat{\mathbf{n}}\right| = \dfrac{\left|\left(\overrightarrow{OA} - \overrightarrow{OP}\right) \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\right|}{\sqrt{2^2 + 1^2 + 2^2}}$

$$=\frac{\left|\begin{pmatrix}3\\-1\\4\end{pmatrix}\cdot\begin{pmatrix}2\\1\\-2\end{pmatrix}-\overrightarrow{OP}\cdot\begin{pmatrix}2\\1\\-2\end{pmatrix}\right|}{\sqrt{2^2+1^2+2^2}}$$ $$=\frac{\left|(6-1-8)-6\right|}{3}=3$$
(ii) $\overline{AB}=\begin{pmatrix}0\\-5\\-2\end{pmatrix}$, $\hat{\underset{\sim}{n}}=\frac{1}{\sqrt{2^2+1^2+(-2)^2}}\begin{pmatrix}2\\1\\-2\end{pmatrix}=\frac{1}{3}\begin{pmatrix}2\\1\\-2\end{pmatrix}$ length of projection $=\left|\overline{AB}\times\hat{\underset{\sim}{n}}\right|=\frac{1}{3}\left|\begin{pmatrix}0\\-5\\-2\end{pmatrix}\times\begin{pmatrix}2\\1\\-2\end{pmatrix}\right|$ $=\frac{1}{3}\left|\begin{pmatrix}12\\-4\\10\end{pmatrix}\right|=\frac{2}{3}\left|\begin{pmatrix}6\\-2\\5\end{pmatrix}\right|=\frac{2}{3}\sqrt{6^2+2^2+5^2}=\frac{2}{3}\sqrt{65}$ units   Alternatively, use $\left|\overline{AB}\bullet\hat{\underset{\sim}{n}}\right|$, followed by Pythagoras thm
(iii) Area of triangle $ABC=\frac{1}{2}(AC)(\frac{2}{3}\sqrt{65})$ $=\frac{1}{2}(3\times 2)(\frac{2}{3}\sqrt{65})=2\sqrt{65}$ units^2
(iv) $\boldsymbol{l}:\boldsymbol{r}=\begin{pmatrix}1\\-4\\5\end{pmatrix}+\lambda\begin{pmatrix}3\\2\\1\end{pmatrix},\lambda\in\Re$, Two vectors // to π_2 are $\begin{pmatrix}1\\-4\\5\end{pmatrix}$ and $\begin{pmatrix}3\\2\\1\end{pmatrix}$,   Normal to π_2 is // to $\begin{pmatrix}1\\-4\\5\end{pmatrix}\times\begin{pmatrix}3\\2\\1\end{pmatrix}=\begin{pmatrix}-14\\14\\14\end{pmatrix}=14\begin{pmatrix}-1\\1\\1\end{pmatrix}$ Equation of π_2: $\boldsymbol{r}\cdot\begin{pmatrix}-1\\1\\1\end{pmatrix}=0$, ie. $-\boldsymbol{x}+\boldsymbol{y}+\boldsymbol{z}=0$ To find line of intersection: $-x+y+z=0$ $2x+y-2z=6$ Using GC,   Equation of line: $\underset{\sim}{r}=\begin{pmatrix}2\\2\\0\end{pmatrix}+\mu\begin{pmatrix}1\\0\\1\end{pmatrix},\mu\in\mathbb{R}$ [Alternatively, Cartesian equation of line: $x-2=z, y=2$]

3 TJC Mid Year Paper 9740/2007/Q7

The line ℓ whose vector equation is $\mathbf{r} = 5\mathbf{i} + 2\mathbf{j} - 3\mathbf{k} + \alpha(7\mathbf{i} - 2\mathbf{j} - 3\mathbf{k})$ passes through the point A with position vector $5\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$. The plane π whose vector equation is

$$\mathbf{r} = -5\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \beta(7\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}) + \gamma(9\mathbf{i} - \mathbf{j} + \mathbf{k})$$

contains the point B with position vector $-5\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

(i) Find the position vector of the point P on the line segment AB such that $AP : PB = 4:1$. [2]

(ii) The plane π_1 contains the line ℓ and the point P.

Write down a vector equation of the line of intersection of π and π_1. [2]

Find the vector equation of π_1 and the angle between π and π_1. [5]

Hence find the ratio of the perpendicular distances from P to the line ℓ and from P to the plane π. [2]

(i)	By ratio theorem, $\overrightarrow{OP} = \frac{1}{5}\left(4\begin{pmatrix}-5\\2\\2\end{pmatrix} + \begin{pmatrix}5\\2\\-3\end{pmatrix}\right) = \begin{pmatrix}-3\\2\\1\end{pmatrix}$
(ii)	π_1 contains AP, hence it contains B. Both π and π_1 // to vector $\begin{pmatrix}7\\-2\\-3\end{pmatrix}$. Hence the line of intersection of π and π_1 is $\underset{\sim}{r} = \begin{pmatrix}-5\\2\\2\end{pmatrix} + \lambda\begin{pmatrix}7\\-2\\-3\end{pmatrix}, \quad \lambda \in \mathbb{R}.$
	$\overrightarrow{AB} = \begin{pmatrix}-5\\2\\2\end{pmatrix} - \begin{pmatrix}5\\2\\-3\end{pmatrix} = 5\begin{pmatrix}-2\\0\\1\end{pmatrix}$ $\therefore$ Eq. of π_1 is $\underset{\sim}{r} = \begin{pmatrix}5\\2\\-3\end{pmatrix} + s\begin{pmatrix}7\\-2\\-3\end{pmatrix} + t\begin{pmatrix}-2\\0\\1\end{pmatrix}$ (or $\underset{\sim}{r}\cdot\begin{pmatrix}2\\1\\4\end{pmatrix} = 0$) normal of π is $\underset{\sim}{n} = \begin{pmatrix}7\\-2\\-3\end{pmatrix} \times \begin{pmatrix}9\\-1\\1\end{pmatrix} = \begin{pmatrix}-5\\-34\\11\end{pmatrix}$

normal of π_1 is $\underset{\sim}{m} = \begin{pmatrix} 7 \\ -2 \\ -3 \end{pmatrix} \times \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ -4 \end{pmatrix}$ $\cos\theta = \dfrac{\begin{pmatrix} -5 \\ -34 \\ 11 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -1 \\ -4 \end{pmatrix}}{\sqrt{1302}\sqrt{21}} = \dfrac{10+34-44}{\sqrt{1302}\sqrt{21}} = 0$ $\therefore$ $\angle$ between π and π_1 is 90°.
$MP : PN = AP{:}PB = 4{:}1$.

4 IJCPrelim9740/2010/1/Q12 modified

A plane π_1 has equation $\mathbf{r}\cdot\begin{pmatrix}2\\-1\\3\end{pmatrix}=9$ and a line ℓ_1 has equation $\mathbf{r}=\begin{pmatrix}3\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}4\\1\\-2\end{pmatrix}$.

(i) Find the coordinates of P, the point of intersection of ℓ_1 and π_1. [3]

Hence, or otherwise, find the shortest distance from point A (3, 0, 0) to π_1. [2]

The equation of a plane π_2 is given as

$$\pi_2 : \quad \mathbf{r}=\begin{pmatrix}1\\1\\1\end{pmatrix}+s\begin{pmatrix}2\\0\\-3\end{pmatrix}+t\begin{pmatrix}2\\2\\1\end{pmatrix}$$

(ii) Find the equation of plane π_2 in the form $\mathbf{r}\bullet\mathbf{n}=d$. Explain why the planes π_1 and π_2 intersect. [4]

(i) $\pi_1 : \quad \mathbf{r}\cdot\begin{pmatrix}2\\1\\-3\end{pmatrix}=9$

$\ell_1 : \quad \mathbf{r}=\begin{pmatrix}3\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}4\\1\\-2\end{pmatrix}$

For point of intersection,

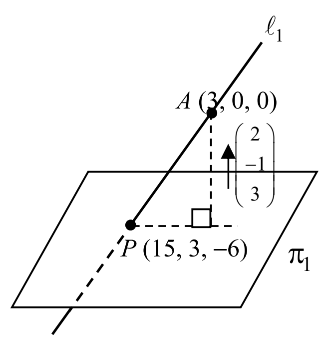
$$\left[\begin{pmatrix}3\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}4\\1\\-2\end{pmatrix}\right]\bullet\begin{pmatrix}2\\-1\\3\end{pmatrix}=9$$

$$\Rightarrow \quad \begin{pmatrix}3\\0\\0\end{pmatrix}\bullet\begin{pmatrix}2\\-1\\3\end{pmatrix}+\lambda\begin{pmatrix}4\\1\\-2\end{pmatrix}\bullet\begin{pmatrix}2\\-1\\3\end{pmatrix}=9$$

$$\Rightarrow \quad \lambda=\frac{9-6}{8-1-6}=3$$

$$\Rightarrow \quad \overrightarrow{OP}=\begin{pmatrix}3\\0\\0\end{pmatrix}+3\begin{pmatrix}4\\1\\-2\end{pmatrix}=\begin{pmatrix}15\\3\\-6\end{pmatrix}$$

Thus, coordinates of the P are $(15,\ 3,-6)$.



$$\overrightarrow{AP} = \begin{pmatrix} 15 \\ 3 \\ -6 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 12 \\ 3 \\ -6 \end{pmatrix}$$

Shortest distance from A to π_1

$$= \frac{\left| \overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right|}$$

$$= \frac{\left| \begin{pmatrix} 12 \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right|}{\sqrt{4+1+9}} = = \frac{|24-3-18|}{\sqrt{14}} = \frac{3}{\sqrt{14}} = \frac{3\sqrt{14}}{14}$$

Alternative solution for shortest distance (to be avoided as it is not necessary to find foot of perpendicular from A to π_1):

Line AN:

$$\underset{\sim}{r} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

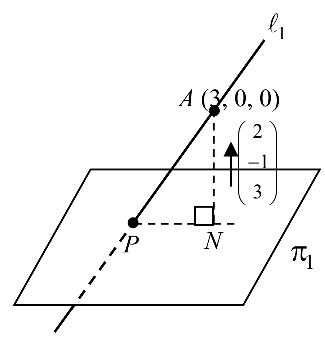
N is the point of intersection of line AN and plane π_1.

$$\left[\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right] \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 9$$

$$\Rightarrow \quad \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 9$$

$$\Rightarrow \quad 6 + (4+1+9)\lambda = 9$$

$$\Rightarrow \quad \lambda = \frac{3}{14}$$



Thus, $\overrightarrow{ON} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \frac{3}{14} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 48 \\ -3 \\ 9 \end{pmatrix}$

$$\overrightarrow{AN} = \frac{1}{14}\begin{pmatrix}48\\-3\\9\end{pmatrix} - \begin{pmatrix}3\\0\\0\end{pmatrix} = \frac{3}{14}\begin{pmatrix}2\\-1\\3\end{pmatrix}$$

Thus, shortest distance AN

$$= \left|\overrightarrow{AN}\right| = \frac{3}{14}\sqrt{4+1+9} = \frac{3\sqrt{14}}{14}$$

(ii) $\pi_1 : \quad \underset{\sim}{r} \cdot \begin{pmatrix}2\\-1\\3\end{pmatrix} = 9$

$\pi_2 : \quad \underset{\sim}{r} = \begin{pmatrix}1\\1\\1\end{pmatrix} + s\begin{pmatrix}2\\0\\-3\end{pmatrix} + t\begin{pmatrix}2\\2\\1\end{pmatrix}$

A normal to plane $\pi_2 = \begin{pmatrix}2\\0\\-3\end{pmatrix} \times \begin{pmatrix}2\\2\\1\end{pmatrix} = \begin{pmatrix}6\\-(8)\\4\end{pmatrix} = 2\begin{pmatrix}3\\-4\\2\end{pmatrix}$

$\pi_2 : \quad \underset{\sim}{r} \cdot \begin{pmatrix}3\\-4\\2\end{pmatrix} = \begin{pmatrix}1\\1\\1\end{pmatrix} \cdot \begin{pmatrix}3\\-4\\2\end{pmatrix} = 3-4+2 = 1$

$\Rightarrow \quad \underset{\sim}{r} \cdot \begin{pmatrix}3\\-4\\2\end{pmatrix} = 1$

Since $\begin{pmatrix}3\\4\\2\end{pmatrix} \neq k\begin{pmatrix}2\\-1\\3\end{pmatrix}$ for any $k \in \mathbb{R}$,

the **normal** of planes π_1 and π_2 are **not parallel** to each other

$\Rightarrow$ The **planes** are **not parallel** to each other.

$\Rightarrow$ The planes will intersect in a line.

5 SRJCPrelim9740/2010/1/Q9

The position vectors of the points A, B, C and D are given as $\mathbf{i}+3\mathbf{j}$, $2\mathbf{j}+4\mathbf{k}$, $\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $4\mathbf{i}+5\mathbf{k}$ respectively.

(i) Find the vector equation of plane π in the form $\mathbf{r}\cdot\mathbf{n}=p$, that contains the points A, B and C. [3]

(ii) Find the foot of the perpendicular of the point D to the plane π. Hence find the shortest distance from the point D to the plane π. [4]

A line l parallel to the vector $\mathbf{j}+\mathbf{k}$ passes through point D and it meets the plane π at the point N.

Find the position vector of the point N and hence find the vector equation of the reflection of line l about the plane π. [5]

(i) $\overrightarrow{AB}=\begin{pmatrix}-1\\-1\\4\end{pmatrix}, \overrightarrow{BC}=\begin{pmatrix}1\\-1\\-3\end{pmatrix} \Rightarrow \overrightarrow{AB}\times\overrightarrow{BC}=\begin{pmatrix}7\\1\\2\end{pmatrix}$

Equation of plane π: $\mathbf{r}\cdot\begin{pmatrix}7\\1\\2\end{pmatrix}=\begin{pmatrix}1\\3\\0\end{pmatrix}\cdot\begin{pmatrix}7\\1\\2\end{pmatrix}=10 \Rightarrow \mathbf{r}\cdot\begin{pmatrix}7\\1\\2\end{pmatrix}=10$

(ii) Let foot of perpendicular from D be F.

Now, $\overrightarrow{OF}=\begin{pmatrix}4\\0\\5\end{pmatrix}+\mu\begin{pmatrix}7\\1\\2\end{pmatrix}$

Since F is on π, $\left[\begin{pmatrix}4\\0\\5\end{pmatrix}+\mu\begin{pmatrix}7\\1\\2\end{pmatrix}\right]\cdot\begin{pmatrix}7\\1\\2\end{pmatrix}=10 \Rightarrow 38+54\mu=10 \Rightarrow \mu=-\frac{14}{27}$

$$\overrightarrow{OF}=\begin{pmatrix}4\\0\\5\end{pmatrix}-\frac{14}{27}\begin{pmatrix}7\\1\\2\end{pmatrix}=\frac{1}{27}\begin{pmatrix}10\\-14\\107\end{pmatrix}$$

Shortest distance from the point D to the plane π,

$$\left|\overrightarrow{DF}\right| = \left|-\frac{14}{27}\begin{pmatrix}7\\1\\2\end{pmatrix}\right|$$

$$= \frac{14}{27}\sqrt{7^2+1+2^2}$$

$$= \frac{14}{27}\sqrt{54} = \frac{14}{9}\sqrt{6}$$

(iii) $l:$ $\quad \mathbf{r} = \begin{pmatrix}4\\0\\5\end{pmatrix} + \lambda\begin{pmatrix}0\\1\\1\end{pmatrix},\ \lambda \in \mathbb{R}$

Thus since $\overrightarrow{ON} = \begin{pmatrix}4\\0\\5\end{pmatrix} + \lambda\begin{pmatrix}0\\1\\1\end{pmatrix}$ lies on plane π,

$$\left[\begin{pmatrix}4\\0\\5\end{pmatrix} + \lambda\begin{pmatrix}0\\1\\1\end{pmatrix}\right]\cdot\begin{pmatrix}7\\1\\2\end{pmatrix} = 10 \quad\Rightarrow\quad 38 + 3\lambda = 10 \quad\Rightarrow\quad \lambda = -\frac{28}{3}$$

Therefore $\overrightarrow{ON} = \begin{pmatrix}4\\-\frac{28}{3}\\-\frac{13}{3}\end{pmatrix}$

Now, $\overrightarrow{OF} = \frac{1}{2}\left(\overrightarrow{OD} + \overrightarrow{OD}'\right) \Rightarrow \overrightarrow{OD}' = \begin{pmatrix}-\frac{88}{27}\\-\frac{28}{27}\\\frac{79}{27}\end{pmatrix}$

Thus $\Rightarrow \overrightarrow{ND}' = \frac{28}{27}\begin{pmatrix}-7\\8\\7\end{pmatrix} \Rightarrow l':\mathbf{r} = \begin{pmatrix}4\\-\frac{28}{3}\\-\frac{13}{3}\end{pmatrix} + \beta\begin{pmatrix}-7\\8\\7\end{pmatrix}$

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(i) Interpret geometrically the vector equation $\mathbf{r} = \mathbf{a} + t\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors and t is a parameter. [2]

(ii) Interpret geometrically the vector equation $\mathbf{r}.\mathbf{n} = d$, where $\mathbf{n}$ is a constant unit vector and d is a constant scalar, stating what d represents. [3]

(iii) Given that $\mathbf{b}.\mathbf{n} \neq 0$, solve the equations $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ and $\mathbf{r}.\mathbf{n} = d$ to find $\mathbf{r}$ in terms of $\mathbf{a}, \mathbf{b}, \mathbf{n}$ and d. Interpret the solution geometrically. [3]

(i) A straight line which passes through a fixed point A with position vector $\mathbf{a}$ and is parallel to a vector $\mathbf{b}$ has the equation $\mathbf{r} = \mathbf{a} + t\mathbf{b}$, where $\mathbf{r}$ represents the position vector of any point on this line.
(ii) A plane that is perpendicular to the vector $\mathbf{n}$, such that the absolute value of d is the distance from origin to plane (We can also describe d as the signed distance). This plane has equation $\mathbf{r}.\mathbf{n} = d$, where $\mathbf{r}$ represents the position vector of any point on this plane.
(iii) $\mathbf{b}.\mathbf{n} \neq 0$ implies that the line is not perpendicular to the normal of the plane, which means that the line is not parallel to the plane. Hence there should be one point of intersection between the line and plane. Substitute the equation of $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ into $\mathbf{r}.\mathbf{n} = d$. $(\mathbf{a} + t\mathbf{b}).\mathbf{n} = d$ $\mathbf{a}.\mathbf{n} + t\mathbf{b}.\mathbf{n} = d$ $t = \dfrac{d - \mathbf{a}.\mathbf{n}}{\mathbf{b}.\mathbf{n}}$ $\mathbf{r} = \mathbf{a} + t\mathbf{b} = \mathbf{a} + \left(\dfrac{d - \mathbf{a}.\mathbf{n}}{\mathbf{b}.\mathbf{n}}\right)\mathbf{b}$ The solution gives the position vector of the point of intersection between the line and plane in part **(i)** and part **(ii)**.

7 NJCMidYear9740/2007/1/Q12 modified

The plane Π_1 has equation $\mathbf{r} \cdot (-\mathbf{i}+2\mathbf{k}) = -4$ and the points A and P have position vectors $4\mathbf{i}$ and $\mathbf{i}+\alpha\mathbf{j}+\mathbf{k}$ respectively, where $\alpha \in \mathbb{R}$.

(i) Show that A lies on Π_1 but P does not. [1]

(ii) Find, in terms of α, the position vector of N, the foot of perpendicular of P on Π_1. [3]

The plane Π_2 contains the points A, P and N.

(iii) Show that the equation of Π_2 is $\mathbf{r} \cdot (2\alpha\mathbf{i}+5\mathbf{j}+\alpha\mathbf{k}) = 8\alpha$ and write down the equation of l, the line of intersection of Π_1 and Π_2. [3]

The plane Π_3 has equation $\mathbf{r} \cdot (\mathbf{i}+\mathbf{j}+2\mathbf{k}) = 4$.

(iv) By considering l, or otherwise, find the value of α for which the three planes intersect in a line. [2]

(i)	$\begin{pmatrix}4\\0\\0\end{pmatrix} \cdot \begin{pmatrix}-1\\0\\2\end{pmatrix} = -4+0+0 = -4$. Hence A lies on Π_1. $\begin{pmatrix}1\\\alpha\\1\end{pmatrix} \cdot \begin{pmatrix}-1\\0\\2\end{pmatrix} = 1 \neq -4$. Hence P is not on Π_1.
(ii)	l_{PN} : $\mathbf{r} = \begin{pmatrix}1\\\alpha\\1\end{pmatrix} + \lambda\begin{pmatrix}-1\\0\\2\end{pmatrix}$ Since N lies on both l_{PN} and Π_1, then we have $\left[\begin{pmatrix}1\\\alpha\\1\end{pmatrix} + \lambda\begin{pmatrix}-1\\0\\2\end{pmatrix}\right] \cdot \begin{pmatrix}-1\\0\\2\end{pmatrix} = -4$ $(-1+2)+\lambda(1+4) = -4$ $\lambda = -1$ Hence $\overrightarrow{ON} = \begin{pmatrix}2\\\alpha\\-1\end{pmatrix}$.
(iii)	Normal of Π_2 $= \overrightarrow{AP} \times \underset{\sim}{n}_1 = \begin{pmatrix}-3\\\alpha\\1\end{pmatrix} \times \begin{pmatrix}-1\\0\\2\end{pmatrix}$

$$= \begin{pmatrix} 2\alpha \\ -(-6+1) \\ \alpha \end{pmatrix} = \begin{pmatrix} 2\alpha \\ 5 \\ \alpha \end{pmatrix}$$ $$\Pi_2 : \quad \mathbf{r} \cdot \begin{pmatrix} 2\alpha \\ 5 \\ \alpha \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2\alpha \\ 5 \\ \alpha \end{pmatrix}$$ $$\mathbf{r} \cdot \begin{pmatrix} 2\alpha \\ 5 \\ \alpha \end{pmatrix} = 8\alpha$$ l is the line AN with equation $\underset{\sim}{r} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ \alpha \\ -1 \end{pmatrix}, \quad \mu \in \mathbb{R}$.
(iv) Since the 3 planes intersect in a line, l must lie on Π_3. Hence we have $$\left[\begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ \alpha \\ -1 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 4 \qquad \text{for all } \mu \in \mathbb{R}$$ $$4 + \mu(-2+\alpha-2) = 4 \qquad \text{for all } \mu \in \mathbb{R}$$ $$\mu(\alpha-4) = 0 \qquad \text{for all } \mu \in \mathbb{R}$$ $$\alpha = 4$$

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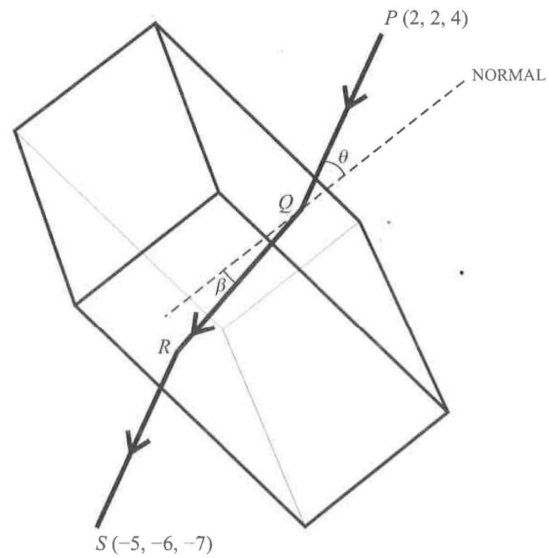

A ray of light passes from air into a material made into a rectangular prism. The ray of light is sent in direction $\begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix}$ from a light source at the point P with coordinates $(2,2,4)$. The prism is placed so that the ray of light passes through the prism, entering at the point Q and emerging at the point R and is picked up by a sensor at point S with coordinates $(-5,-6,-7)$. The acute angle between PQ and the normal to the top of the prism at Q is θ and the acute angle between QR and the same normal is β (see diagram).

It is given that the top of the prism is a part of the plane $x+y+z=1$, and that the base of the prism is a part of the plane $x+y+z=-9$. It is also given that the ray of light along PQ is parallel to the ray of light along RS so that P, Q, R and S lie in the same plane.

(i) Find the exact coordinates of Q and R. [5]

(ii) Find the values of $\cos\theta$ and $\cos\beta$. [3]

(iii) Find the thickness of the prism measured in the direction of the normal at Q. [3]

Snell's law states that $\sin\theta = k\sin\beta$, where k is a constant called the refractive index.

(iv) Find k for the material of this prism. [1]

(v) What can be said about the value of k for a material for which $\beta > \theta$? [1]

(i)	$\overrightarrow{PQ}=\lambda\begin{pmatrix}-2\\-3\\-6\end{pmatrix}, \lambda>0$ and Q lies on $\underset{\sim}{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=1$ $\overrightarrow{OQ}=\lambda\begin{pmatrix}-2\\-3\\-6\end{pmatrix}+\begin{pmatrix}2\\2\\4\end{pmatrix}$ $\overrightarrow{OQ}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=1\Rightarrow -2\lambda+2-3\lambda+2-6\lambda+4=1\Rightarrow \lambda=\frac{7}{11}$ Q is $\left(2-\frac{14}{11},2-\frac{21}{11},4-\frac{42}{11}\right)=\left(\frac{8}{11},\frac{1}{11},\frac{2}{11}\right)$ $\overrightarrow{RS}=\mu\begin{pmatrix}-2\\-3\\-6\end{pmatrix}, \mu>0$ and R lies on $\underset{\sim}{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=-9$ $\overrightarrow{OR}=\begin{pmatrix}-5\\-6\\-7\end{pmatrix}-\mu\begin{pmatrix}-2\\-3\\-6\end{pmatrix}$ $\overrightarrow{OR}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=-9\Rightarrow -5+2\mu-6+3\mu-7+6\mu=-9\Rightarrow \mu=\frac{9}{11}$ R is $\left(-5+\frac{18}{11},-6+\frac{27}{11},-7+\frac{54}{11}\right)=\left(-\frac{37}{11},-\frac{39}{11},-\frac{23}{11}\right)$	M1 valid method to find Q or R A1 value of λ A1 coordinates of Q A1 value of μ A1 coordinates of R
(ii)	$\cos\theta=\dfrac{\left\lvert\begin{pmatrix}-2\\-3\\-6\end{pmatrix}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}\right\rvert}{\sqrt{4+9+36}\sqrt{3}}=\dfrac{11}{7\sqrt{3}}$ $\overrightarrow{RQ}=\frac{1}{11}\begin{pmatrix}8\\1\\2\end{pmatrix}-\frac{1}{11}\begin{pmatrix}-37\\-39\\-23\end{pmatrix}=\frac{1}{11}\begin{pmatrix}45\\40\\25\end{pmatrix}$ $\cos\beta=\dfrac{\left\lvert\begin{pmatrix}45\\40\\25\end{pmatrix}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}\right\rvert}{\sqrt{45^2+40^2+25^2}\sqrt{3}}=\dfrac{110}{\sqrt{12750}}=\dfrac{22}{\sqrt{510}}$	M1 A1 $\cos\theta$ A1 $\cos\beta$

(iii)	Thickness of prism is $\dfrac{\left\lvert \overrightarrow{RQ} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\rvert}{\sqrt{3}} = \dfrac{\frac{1}{11}(110)}{\sqrt{3}} = \dfrac{10}{\sqrt{3}}$	M1 length of projection A1 correct expression A1
(iv)	$k = \dfrac{\sin\theta}{\sin\beta} = \sqrt{\dfrac{1-\cos^2\theta}{1-\cos^2\beta}} = \sqrt{\dfrac{1-\left(\frac{11}{7\sqrt{3}}\right)^2}{1-\left(\frac{22}{\sqrt{510}}\right)^2}} = \dfrac{\sqrt{170}}{7} = 1.86$ (3 s.f.)	B1
(v)	Since $0^\circ < \theta < \beta < 90^\circ$, we have $0 < \sin\theta < \sin\beta < 1$. Thus, $0 < \dfrac{\sin\theta}{\sin\beta} < 1$, that is, $0 < k < 1$.	B1

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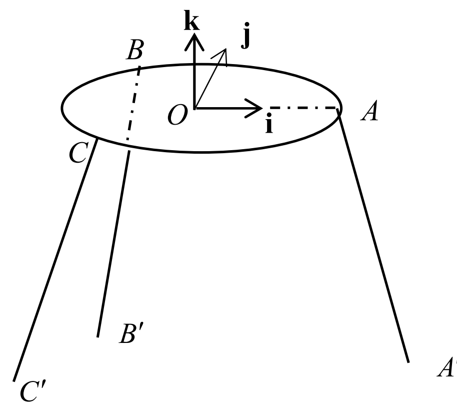
The diagram shows a circular table, with centre O, and three legs AA', BB' and CC', attached at the points A, B and C to the table. The lengths of the legs are adjusted so that, although the table is standing on a sloping floor, the plane ABC is horizontal. Perpendicular vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are defined with $\mathbf{i}$ along $\overrightarrow{OA}$, and $\mathbf{k}$ vertically upwards.



The vectors $\overrightarrow{AA'}$, $\overrightarrow{BB'}$ and $\overrightarrow{CC'}$ are parallel to $5\mathbf{i}-12\mathbf{k}$, $-3\mathbf{i}+4\mathbf{j}-12\mathbf{k}$ and $-3\mathbf{i}-4\mathbf{j}-12\mathbf{k}$ respectively.

(i) Show that all three legs are inclined at the same angle to the vertical. [3]

(ii) The plane $A'B'C'$ has equation $x-10z=130$. Find the inclination of this plane to the horizontal, and the perpendicular distance from O to this plane. [4]

(iii) Referred to O as the origin, the position vectors of A, B and C are $5\mathbf{i}$, $-3\mathbf{i}+4\mathbf{j}$ and $-3\mathbf{i}-4\mathbf{j}$ respectively. Find the lengths of the legs AA', BB' and CC'. [7]

(i)	Angle that AA' makes with the vertical = $\cos^{-1}\dfrac{\left\|\begin{pmatrix}5\\0\\-12\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}\right\|}{\sqrt{25+144}}=\cos^{-1}\dfrac{12}{13}$ Angle that BB' makes with the vertical = $\cos^{-1}\dfrac{\left\|\begin{pmatrix}-3\\4\\-12\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}\right\|}{\sqrt{9+16+144}}=\cos^{-1}\dfrac{12}{13}$ Angle that CC' makes with the vertical = $\cos^{-1}\dfrac{\left\|\begin{pmatrix}-3\\-4\\-12\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}\right\|}{\sqrt{9+16+144}}=\cos^{-1}\dfrac{12}{13}$ Therefore, all three legs are inclined at the same angle to the vertical.
(ii)	Inclination of plane $A'B'C'$ to the horizontal = $\cos^{-1}\dfrac{\left\|\begin{pmatrix}0\\0\\1\end{pmatrix}\bullet\begin{pmatrix}1\\0\\-10\end{pmatrix}\right\|}{\sqrt{101}}=5.7^{o}$

$\pi_{A'B'C'}: \quad \mathbf{r}\bullet\begin{pmatrix}1\\0\\-10\end{pmatrix}=130$

(130, 0, 0) lies in $\pi_{A'B'C'}$

Perpendicular distance from O to plane $A'B'C'$ = $\left|\begin{pmatrix}130\\0\\0\end{pmatrix}\bullet\begin{pmatrix}1\\0\\-10\end{pmatrix}\right|\frac{1}{\sqrt{101}}=\frac{130}{\sqrt{101}}$

(iii) $l_{AA'}: \quad \mathbf{r}=\begin{pmatrix}5\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}5\\0\\-12\end{pmatrix}, \lambda\in\mathbb{R}$

$$\left[\begin{pmatrix}5\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}5\\0\\-12\end{pmatrix}\right]\bullet\begin{pmatrix}1\\0\\-10\end{pmatrix}=130$$

$$5+\lambda(5+120)=130$$

$$\lambda=1$$

$$\overrightarrow{AA'}=\begin{pmatrix}5\\0\\-12\end{pmatrix}$$

$$|AA'|=\sqrt{25+144}=13$$

$l_{BB'}: \quad \mathbf{r}=\begin{pmatrix}-3\\4\\0\end{pmatrix}+\lambda\begin{pmatrix}-3\\4\\-12\end{pmatrix}, \lambda\in\mathbb{R}$

$$\left[\begin{pmatrix}-3\\4\\0\end{pmatrix}+\lambda\begin{pmatrix}-3\\4\\-12\end{pmatrix}\right]\bullet\begin{pmatrix}1\\0\\-10\end{pmatrix}=130$$

$$-3+\lambda(-3+120)=130$$

$$\lambda=\frac{133}{117}$$

$$\overrightarrow{BB'}=\frac{133}{117}\begin{pmatrix}-3\\4\\-12\end{pmatrix}$$

$$|BB'|=\frac{133}{117}\sqrt{9+16+144}=\frac{133}{9}$$

$$l_{CC'}: \quad \mathbf{r} = \begin{pmatrix} -3 \\ -4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -4 \\ -12 \end{pmatrix}, \lambda \in \mathbb{R}$$

$$\left[\begin{pmatrix} -3 \\ -4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -4 \\ -12 \end{pmatrix} \right] \bullet \begin{pmatrix} 1 \\ 0 \\ -10 \end{pmatrix} = 130$$

$$-3 + \lambda(-3 + 120) = 130$$

$$\lambda = \frac{133}{117}$$

$$\overrightarrow{CC'} = \frac{133}{117} \begin{pmatrix} -3 \\ -4 \\ -12 \end{pmatrix}$$

$$|CC'| = \frac{133}{117}\sqrt{9+16+144} = \frac{133}{9}$$

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The plane π passes through the origin O and is perpendicular to $2\mathbf{i}+2\mathbf{j}+\mathbf{k}$. The line L has Cartesian equation $\frac{x-4}{1} = \frac{y-8}{3} = \frac{z-4}{2}$. The points A and B have position vectors $\mathbf{i}+2\mathbf{j}-3\mathbf{k}$ and $3\mathbf{i}+5\mathbf{j}+2\mathbf{k}$ respectively.

(i) State a Cartesian equation for π. [1]

(ii) The point C on L has position vector $\mathbf{c}$, where $\mathbf{c}\cdot\mathbf{k} = 0$. Find $\mathbf{c}$, and hence state an alternative equation for L of the form $\frac{x-p}{l} = \frac{y-q}{m} = \frac{z}{n}$, giving numerical values for p, q, l, m, and n. [4]

(iii) Find the exact value of the length of perpendicular from A to L. [2]

(iv) Find, in either order, the exact value of the length of projection of $\overrightarrow{AB}$ onto

(a) the normal to π, [2]

(b) the plane π. [2]

(i)	$2x+2y+z=0$
(ii)	$L: \quad \mathbf{r} = \begin{pmatrix} 4 \\ 8 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R}$

$$\left[\begin{pmatrix}4\\8\\4\end{pmatrix}+\lambda\begin{pmatrix}1\\3\\2\end{pmatrix}\right]\bullet\begin{pmatrix}0\\0\\1\end{pmatrix}=0$$

$$4+2\lambda=0$$

$$\lambda=-2$$

$$\mathbf{c}=\begin{pmatrix}2\\2\\0\end{pmatrix}$$

$$L: \quad \frac{x-2}{1}=\frac{y-2}{3}=\frac{z}{2}$$

Alternatively,

c.**k** = 0 implies z = 0. Substituting into the cartesian equation of L we immediately get $x = y =$ 2. Similarly,

$$L: \quad \frac{x-2}{1}=\frac{y-2}{3}=\frac{z}{2},$$

(iii) Length of perpendicular from A to L = $\left|\overrightarrow{AC}\times\frac{1}{\sqrt{1+9+4}}\begin{pmatrix}1\\3\\2\end{pmatrix}\right|$

$$=\left|\frac{1}{\sqrt{14}}\begin{pmatrix}1\\0\\3\end{pmatrix}\times\begin{pmatrix}1\\3\\2\end{pmatrix}\right|$$

$$=\left|\frac{1}{\sqrt{14}}\begin{pmatrix}9\\1\\3\end{pmatrix}\right|=\sqrt{\frac{91}{14}}=\sqrt{\frac{13}{2}}$$

(iv) $\overrightarrow{AB}=\begin{pmatrix}2\\3\\5\end{pmatrix}$

(a) Length of projection of $\overrightarrow{AB}$ onto normal of π is $\left|\begin{pmatrix}2\\3\\5\end{pmatrix}\bullet\frac{1}{\sqrt{4+4+1}}\begin{pmatrix}2\\2\\1\end{pmatrix}\right|=5$

(b) Length of projection of $\overrightarrow{AB}$ onto plane π is

Method 1: $\sqrt{AB^2 - 5^2} = \sqrt{2^2 + 3^2} = \sqrt{13}$

Method 2: $\left|\begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} \times \frac{1}{\sqrt{4+4+1}}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}\right| = \frac{1}{3}\left|\begin{pmatrix} -7 \\ 8 \\ -2 \end{pmatrix}\right| = \sqrt{13}$

THE END